

PRECIPITATION REACTIONS

When two soluble ionic solutions are mixed it is possible that one of the new combinations of ions produces an ***insoluble salt***. These types of reactions are known as ***precipitation reactions***.

To predict whether a precipitate will form, a table of ***solubility rules*** is used.

SOLUBLE

All ***Group I*** cations (*lithium, sodium, potassium ions*), *ammonium ion*.

All *nitrates* and *acetates*.

All *chlorides, bromides, iodides* except *silver, lead (II) + mercury ions*.

All *sulphates* except *barium, lead (II) + calcium ions*.

INSOLUBLE

All *oxides* and *hydroxides* except ***Group I*** (*lithium, sodium, potassium ions*),
ammonium + barium ions.

All *sulphides* except ***Group I*** (*lithium, sodium, potassium ions*), *ammonium ion*.

All *carbonates, phosphates* and *sulphites* except ***Group I*** (*lithium, sodium, potassium ions*), *ammonium ion*.

Using The Solubility Rules

Barium nitrate and sodium sulphate solutions are mixed. Will a precipitate form?
If yes, write a molecular and net ionic equation to represent the reaction.

1. ***Write down the name for each solution being mixed.*** They can both be indicated as (aq) because they are soluble:

barium nitrate _(aq) + sodium sulphate _(aq)

2. ***Look at the two possible new combinations of ions by swapping the names of the metal ions. Check out the solubility table to see if an insoluble substance is produced.***

The solubility rules indicate that, of the two possible products ***barium sulphate*** and ***sodium nitrate***, the barium sulphate is insoluble.

3. ***Write out an equation that identifies the insoluble substance formed with an (s).*** The other possible combination can be represented with an (aq) to indicate that it remains in solution and is soluble.

barium nitrate _(aq) + sodium sulphate _(aq) → barium sulphate _(s) + sodium nitrate _(aq)

Examples

Write word equations for the following mixtures of solutions. If no precipitate forms, write "no reaction".

- (a) sodium phosphate + aluminium nitrate
- (b) copper (II) chloride + silver nitrate
- (c) ammonium sulphide + potassium bromide
- (d) calcium acetate + sodium carbonate
- (e) potassium hydroxide + lead (II) nitrate
- (f) calcium nitrate + sodium sulphate
- (g) sodium sulphite + potassium phosphate